INCH-POUND

MIL-T-83818 (84)

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TRUCK, LIFT, FORK, 4,000 AND 6,200 POUND CAPACITY FRONT/SIDE LOADING, INTERNAL COMBUSTION ENGINE

This specification is approved for use within WR-ALC, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for a front/side loading, sit-down rider, internal combustion powered, counterbalanced forklift truck [hereinafter called truck). The truck shall have a telescopic mast, front wheel drive, power steering, hydraulic brakes, solid pneumatic tires, and overhead guard.

1.2 Classification. Trucks shall be of the following types as specified (see 6.2):

Type 1: 4,000 pound capacity at 24 inch load center.

Type 2: 6,200 pound capacity at 24 inch load center.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified, herein (see 6.2).

SPECIFICATIONS

FEDERAL

GG-P-455	Plates and Foils, Photographic (Photosensitive anodized aluminum)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMDETB, Robins AFB, GA 31098-5609, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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MILITARY

MIL-P-514	Plates, Identification, instruction, and Marking Blank
MIL-A-11755	Antifreeze, Arctic Type
MIL-A-46153	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base
MIL-C-83286	Coating, Urethane, Aliphatic Iscocyanate for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-297	Rustproofing of Commercial (Nontactical) Vehicles
FED-STD-595	Color

MILITARY

MIL-STD-162	Materials Handing Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Process
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-248	Welding and Brazing Procedures and Performance Qualification
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-889	Dissimilar Metals
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1474	Noise Limits for Army Material
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 1911 1-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1 a

(Application for copies should be addressed to the Superintendent of Documents, Government Printing office, Washington, D.C. 24401.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1910 Occupational Safety and Health Administration (OSHA)

(Application for copies should be addressed to a DOL Office of the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402.)

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DEPARTMENT OF THE AIR FORCE

AFSC DH 1-11 Air Transportability

(Applications for copies should be addressed to HQ AFLC, ASD/ENECA, Wright Patterson AFB OH 45433.}

U.S. Air Force Drawing No. 8838302

(Application for copies should be addressed to WR-ALC/MMDE, Robins AFB GA 31098.)

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Carrier Safety Regulation 393.77(c)(7)

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, D.C. 20591.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

40 CFR, Part 86, Subpart B

Emission Regulation for 1977 and Later Model Year New Light Duty Vehicles and New Light Duty Trucks; Test Procedures (particulate tunnel and sampling system) pp 402-465, 1 July 1983.

40 CFR, Parts 81 to 99, Subparts D and I

Protection of Environment.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

056.1 -Safety Standards for Powered Industrial Trucks

ANSI/UL 558 -Standards for Safety for Internal Combustion Engine

B56.11.3 -Load Handling Symbols for Powered Industrial Trucks

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951- Standard Practice for Commercial Packaging.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATO STANDARD DIESEL AND SPARK IGNITION ENGINES LABORATORY TEST

AEP-5 Test Procedure

(Application for copies should be addressed to U.S. Army Tank and Automotive Command, ATTN: AMSTA-RGRD, Warren, Ml 48397 -5000.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Standards and Recommended Practices

J185	Access Systems for Off-Road Machines, Recommended Practice
J209	instrument F-ace Design and Location for Construction and Industrial Equipment, Recommended Practice
J514	Hydraulic Tube Fittings, Standard
J516	Hydraulic t-lose Fittings, Standard
J517	Hydraulic Hose, Standard
J518	Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type, Standard
J524	Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring, Standard
J525	Welded and Cold Drawn Low Carbon Steel Tubing Annealed for Bending and Flaring, Standard
J534	Lubrication Fittings, Standard
J593	Backup Lamps (Reversing Lamps), Standard
J614	Engine and Transmission Dipstick Marking, Recommended Practice
J753	Lubrication Chart - Construction and Industrial Machinery, Recommended Practice
J759	Lighting Identification Code, Recommended Practice
J873	Human Physical Dimensions, Recommended Practice
J898	Control Locations for Off-Road Work Machines, Standard
J899	Operator's Seat Dimensions for Off-Road, Self-Propelled Work Machines, Standard
J925	Minimum Service Access Dimensions for Off-Road Machines, Standard
J931	Hydraulic Power Circuit Filtration, Information Report
J1176	External Leakage Classifications for Hydraulic Systems, Recommended Practice
J1362	Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines, Recommended Practice

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM Association, INC. (TRA)

Tire and Rim Association Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

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UNDERWRITER'S LABORATORIES, INC.

UL 558 Standard for Internal Combustion Engine Powered Industrial Trucks

(Application for copies should be addressed to UL, Publications Stock, 333 Pfingsten Road, Northbrook IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These -documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall have a diesel, internal combustion engine, or if otherwise specified (see 6.2), the engine shall be a gasoline or liquid petroleum gas (LPG), or dual fuel (gasoline and LPG) type. The truck shall be a sit-down rider, counterbalanced, front/side loading truck with front wheel drive, and rear wheel steer. The truck shall be the latest model of the manufacturer's standard commercial product and shall have been marketed and in use for a minimum of two years preceding the solicitation for procurement. The truck shall be equipped with instruments, components, and accessories required for the operation of the truck. The truck shall be complete with all components that are standard with the contractor's products, whether stipulated herein or not, together with such accessories as may be specified herein.

3.2 First article. The contractor shall furnish one complete truck for First Article examination and testing in accordance with 4.4. Examination and tests shall be conducted by the Contractor at his facility or at another Government approved location. When specified (see 6.2), another truck for the User Test (see 4.9) shall be furnished. The truck for the User Test shall be subjected to examination (see 4.4.1) and production run-in (see 4.7) by the Contractor before being shipped to the User Test site. The truck for the User Test shall be equipped with Type A winterization (see 3.39.1) if any units on the contract require this item. All Testing shall be observed and approved by Government representatives. The approved First Article truck(s) shall be rehabilitated to a new condition and delivered as specified in the contract.

3.3 Materials. Shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the trucks may be newly fabricated from recovered materials to the maximum extent practicable, provided the trucks produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck.

3.3.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals and methods of protecting them are defined and described in MIL-STD-889.

3.3.3 Fasteners. All fasteners of class 5 or higher shall meet the quality assurance requirements of U.S. Air Force Drawing No. 8838302.

3.3.4 Rustproofing. When specified (see 6.2), the following areas of the truck shall be treated in accordance with FED-STD-297 for tropical rustproofing:

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a. Hood

b. Roof

c. Doors

cf. Pillars

"e. Cab interior

f. Other body requirements (cab only)

g. Applicable areas specified for tropical rustproofing (see FED-STD-297, paragraph 5.8)

3.3.5 Drain holes. Structures shall be designed to eliminate pockets where debris and water might accumulate. All structural configurations shall allow water run-off, or shall have adequate sized and placed drain holes.

3.4 Pivot and shift assembly. Pivot and shift assembly shall be provided to allow 90 degrees minimum pivoting of load and full sideshift action to facilitate transfer of pallets (size 48 inches deep by 48 inches wide) between the truck and storage position. Pivot and shift shall be accomplished hydraulically. Pivot shall be 90 degrees plus or minus 3 degrees. With the forks pivoted 90 degrees and shifted to the extreme lift or right when transversing to the right or left the transverse motion shall be such that the vertical, load carrying, face of the fork extends 1 inch minimum beyond the right or left side of the truck. With fork in the front handling position, sideshift shall provide for a minimum movement of 12 inches across the width of the truck. With forks in the sidehandling position and sideshifted, the truck shall be capable of carrying 16 inch wide loads on the deck of the truck within the planned outline of the truck.

3.5 Operation. Truck shall operate over paved surfaces and will be used for transporting long loads and pallet loads. Trucks shall pick Up and handle long loads, at least 20 feet long with a maximum width of 30 inches, in an aisle not more than 83 inches wide. Truck shall select and replenish pallet loads in a pallet storage rack area with 64 inch wide clear operating aisles (pallet face to pallet face). The truck shall also load and unload long loads and pallet loads onto and off of over-the-road flatbed trailers from the side and in and out of bulk storage areas from the front loading and or side loading positions. Type 1 trucks shall be capable of driving onto trailers and trailer vans to load and unload.

3.5.1 Operating temperature. The forklift shall operate as specified herein within 15 minutes after engine start in any ambient temperature in the range of minus 25 degrees F to plus 120 degrees F when tested in accordance with 4.6.34 of this specification. Engine oil, transmission fluid, and drive line fluids and lubricant temperature shall not exceed 250" F during operations and testing specified herein.

3.6 Design. The truck shall be capable of safely handling the rated capacity load at all lift heights, mast pivot and side shift positions specified herein. Trucks covered by this document and components incorporated therein shall conform to accepted commercial design practices. Trucks shall be constructed and equipped to meet all requirements specified herein. Component parts of the unit need not be products of the same manufacturer. However, the manufacturer shall be so established that prompt and continuing service and delivery of repair parts will be assured. The truck shall be equipped with ail other components and parts not specifically mentioned but necessary to provide a functional machine and shall conform in quality to that normally provided to the commercial industry.

3.7 Safety. The trucks shall conform to the applicable requirement of ANSI 856.1 and ANSI/UL 558, MIL-STD-1474, and OSHA in effect at the time of manufacture and other requirements specified herein. Truck shall be ANSI/UL 558 Type GS, DS, LPGS, or GS/LPGS as applicable, and the individual trucks shall be labeled with safety rating in accordance with ANSI/UL 558.

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* 3.7.1 Seatbelt and operator's restraint. An operator's seat belt conforming to SAE J386, Type One, shall be installed. Additional restraint devices (e. g., winged seats, enclosures, etc.) designed to ensure the operator's body remains entirely within the protection of the frame and overhead guard in the event of tipover or lateral roll will also be provided. The restraint system will not interfere with operator access or mobility or the vehicle's operation. A warning decal will be provided advising of the hazards of tipover and the importance of using the restraint system. The installation of the lap belts alone will not be considered as adequate occupant restraint. The manufacturer will provide evidence of the suitability and effectiveness of any provided occupant restraint systems.

3.7.2 Fire extinguisher. A 2.5 pound (minimum) Halon fire extinguisher shall be mounted in an easily accessible location.

3.7.3 Horn. There shall be an electric horn with a pushbutton mounted in the center of the steering wheel.

3.8 Structure.

3.8.1 Chassis and frame. The frame and related structure shall be fabricated from steel and steel castings and shall be capable of withstanding 300 percent of rated load without permanent deformation. Frame members, bracing, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts and shall be capable of withstanding, without permanent deformation, stresses induced by capacity loads. The frame and related structure shall not crack, deform, or fail under the testing specified in Section 4 of this document.

3.8.2 Operator's overhead guard. An overhead guard that meets all requirements of ANSI/B56.1, including falling object protective structure (FOPS) and vertical clearance between operator's seat and the guard shall be furnished.

3.8.3 Width. The overall width of the truck, at the widest point, shall not exceed 45 inches for the Type 1 truck and 51 inches for the Type 2 truck,

3.8.4 Height. The overall truck height, with mast collapsed, shall not exceed 88 inches

3.8.5 Chassis. The chassis shall be capable of supporting all imposed loads when operated and tested as specified herein. Provisions shall be made for immediate and easy access to all working parts requiring inspections and maintenance. Covers shall be secured with quick opening fasteners removable by hand or with ordinary small hand tools.

3.8.6 Pivot and shift asembly. The pivot/shift assembly shall support a rated load at 24 inch load center and shall function as specified in 3.4.

3.9 Human factors. The characteristics of the forklift shall provide for operation and maintenance by personnel ranging from the small person clothed through the large person arctic clothed in accordance with SAE J185, J833, and SAE J925 as specified herein.

3.10 Maintainability. Provision shall be made for adjustment, servicing, and replacement of all electrical assemblies and components, hydraulic system components, battery, wearing parts of lift and tilt mechanism, brakes and components, wear parts of steering assembly, tires, wheels, lights, and horn. If hand access openings are required to perform maintenance, the edges of each opening shall be smooth and the opening shall 'be provided with a removable or hinged cover. Dimensions of access openings shall be in accordance with SAE J925 for mittened hand. There shall be no interference with the servicing or drawing of oils and lubricants to or from any assembly or component by frame members or other obstructions. Oil and lubricants shall be capable of being drained without requiring the use of flexible connections. All drains shall provide for complete drainage. Engine, transmission, and hydraulic dipsticks shall be in accordance with SAE J614.

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3.10.1 Maintenance operation. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified using common tools and, if necessary, the special tools furnished with the truck.

a. Remove, replace and adjust all engine driven belts -- 0.5 hour.

t). Remove and replace alternator -- 0.5 hour.

c. Remove and replace all filters, screens and strainers in all hydraulic systems --0.75 hour.

d. Remove and replace fuel filter elements -- 0.5 hour.

e. Drain engine lubricating oil, remove and replace oil filter elements and refill crankcase -- 0.75 hour.

f. Remove and replace engine coolant system hoses -- 0.5 hour.

g. Disconnect battery cables, remove and replace batteries and reconnect battery cables -- 0.5 hour.

h. Drain transmission oil, remove and replace all filter elements and refill transmission --0.75 hour.

i. Remove and replace starter --0.75 hour.

j. Bleed and adjust brakes, refill master cylinder (two men) --0.75 hour.

k. Remove and replace headlight, floodlight and tail light bulbs --0.25 hour each.

1. Remove air cleaner, replace element, reinstall air cleaner --0.25 hour.

m Lubricate all lubrication fittings with 2 strokes per fitting using grease gun and extension -- 0.5 hour.

All fluids shall be able to drain to the ground without contacting any truck structure or components. Hoses may be provided if necessary to meet this requirement.

3.10.2 Special tools. Any special tools (tools that are peculiar to this truck) that are needed to perform scheduled or unscheduled maintenance shall be provided by the contractor. One set of special tools shall be supplied with each truck.

3.10.3 Lubrication. Means shall be provided for all moving parts that require lubrication, except where sealed permanent lubrication is provided. Each forklift shall be serviced and run-in with lubricants standard to the contractor's commercial practice. A lubrication data plate in accordance with SAE J753 shall be mounted on the forklift and shall identify ail lubricants and appropriate temperature ranges to be utilized when servicing the forklift. The data plate shall be made of corrosion resistant metal. Grease lubrication fittings shall conform to SAE J534.

3.10.4 Servicing. It shall be possible to reach the dipsticks or other indicators for all operational fluid levels without removing any panels from the truck. If the indicators are behind panels, the panels must open without the need for tools and they must be hinged so that they stay on the truck. The engine hood shall be capable of being opened by one man using one hand and shall be securely held open by a counterbalance or positive locking device.

3.11 Transportability. The forklifts shall be capable of being transported by water, highway and rail without significant restrictions. The trucks shall be air transportable in C-130, C-141 and C-5 aircraft without removal of any components (including cap, if so equipped). The truck shall meet all requirements of MIL-STD-1791 and AFSC Design Handbook DH 1-11.

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3.12 Engine. The truck shall be powered by a standard commercial engine. Horsepower and torque characteristics shall be sufficient to provide the performance specified herein, and shall be at a RPM which will assure performance with an adequate margin of safety.

3.12.1 Diesel engine. Emergency shutdown provisions shall be furnished when recommended by the engine manufacturer. The engine shall operate on diesel fuel, or JP-8 conforming to MIL-T-83133 without detrimental effect on the engine. When operating on JP-8 the truck shall meet the performance requirements of Section 4 of this document. The engine shall be certified that it has successfully completed NATO AEP-5 test procedures (400 hour dynamometer test} using only JP-8.

3.13 Fuel system.

3.13.1 Gasoline or diesel engine fuel system. A fuel filter shall be provided and shall be accessible without removal of other components, from above or from the side. Diesel fuel system shall include a water separator with provisions for draining sediment. The capacity of the fuel tank shall ensure not less than 8 hours of continuous operation during the first article of endurance testing (see 4.6.33). The tank shall have a filler cap that meets UL 558 for GS and DS rated trucks. The tank filling system shall accept a fuel nozzle with a 2.0 inch outside diameter with clearance for air to escape during refueling. The fuel tank shall be equipped with a drain plug at the lowest point of the tank. The drain plug shall be removable with hand tools without having to remove any other component, The drain plug shall be located such that fuel can be completely drained. Fuel type shall be stenciled near fuel cap.

3.13.2 Liquefied petroleum gas engine fuel system. A fuel filter shall be provided and shall be accessible from above or from the side without removal of other components. The capacity of the fuel container shall ensure not less than 8 hours of continuous operation during first article endurance testing specified in 4.6.33.

3.14 Engine accessories for gasoline, liquefied petroleum gas (LPG) and diesel ermines.

3.14.1 Cold weather starting system. When diesel engine is specified, integral starting aids shall be provided to enable engine to be started" under cold weather conditions of 32° F to -25° F. When cold weather start system using ether is provided, a measured shot system shall be furnished that is not functional before cranking engine and not functional after engine has started. A fuel heater, warmed by engine coolant shall be provided. LPG and gasoline engines shall start without starting aids at temperatures as low as -25° F.

3.14.2 Crankcase ventilation system. A positive crankcase ventilation system shall be furnished that shall prevent crankcase vapor from entering the atmosphere.

3.14.3 Alternator. An alternator shall be provided and shall provide charge at engine idle speed and shall maintain battery charge at full electric load. A regulator which may be integral with the alternator shall be furnished.

3.14.4 Starting system. A device shall be furnished to prevent re-energizing the starter motor after the engine has been started. Trucks shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor except when the transmission is in the neutral position.

3.14.5 Air induction system. A dry, replaceable cartridge type air cleaner shall be furnished unless prohibited by ANSI/UL 558 for DS rated trucks. The cleaner shall be a multi-stage air cleaner and shall include a dust scavenger. It shall be accessible without removal of other components when engine cowl or hood is open. A restriction indicator shall be furnished to indicate when the replaceable filter element shall be changed.

* 3.14.6 Exhaust system. Muffler and exhaust piping shall conform to the requirements of ANSI/UL 558. Unless a turbocharged diesel engine is furnished, a spark arrest shall be provided for diesel engines which meets requirements of Forest Service Standard 5100-1a. The spark arrestor shall be sized to meet

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requirements for engine exhaust flow at rated engine speed in accordance with Forest Service Standard 5100-1 a. A means shall be provided to clean out spark arrestor without removal of any components. .-Exhaust discharges shall be directed away from the operator.

3.15 Transmission. The transmission shall be in combination with a torque converter. It shall be a one-speed with a hydraulic direction reverser to provide direction changes without stopping, or a multi-speed powershift type. If applicable, shifting between gear ratio shall be accomplished by selective engagement of multiple-disk clutches which shall be actuated by annular hydraulic pistons. Actuation shall be controlled by a hydraulic control valve body mounted on or in the transmission housing. The transmission shall provide for shifting under the full engine power through all successive gear ratios in each direction. The transmission shall provide controlled application of the clutches or provide smooth shifting, or smooth transition when changing directions, without slippage that could damage the clutches. Oil pressure for actuation of the transmission clutches, for torque converter supply, for and transmission lubrication shall be supplied by a mechanically driven pump. The transmission and torque converter fluid shall be filtered by full flow filter(s), having spin-on type replaceable filters. The transmission shall provide for positive inching control or declutch control of the truck (throughout the entire range of engine speed) in the forward and reverse directions. The inching or declutch control permit tilting and full lifting speeds with rated load while the transmission control is in forward or reverse positions without vehicle movement.

3.16 Directional controls. Selective forward and reverse directional controls actuated by the operator's toot are not acceptable.

3.17 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the load handling system and shall not include transmission operation. The hydraulic system fluid shall comply with MIL-H-83282, and all hydraulic system components shall be compatible with this fluid. The hydraulic lift system shall be fail safe as defined in 6.4. A pipe tap or straight thread O-ring boss test point for measuring pressure shall be located at the outlet of each pump or pressure inlet of each control valve. Hydraulic hose(s) shall have a working pressure which is greater than the hydraulic system maximum relief valve setting. A system filter shall be furnished in either the suction or return line. The system filter shall meet the requirements of SAE J931a. Hose reels shall be mounted such that they are within the plan outline of the truck. Positive means shall be furnished to prevent cavitation of the tilt cylinders.

3.17.1 Hydraulic tubing hoses and fittings. All hydraulic tubing, hoses and fittings shall meet the requirements of the applicable SAE recommended practices paragraphs J514, J516, J517, J518, J524, and J525 and shall be routed to preclude kinking, chafing and twisting.

3.17.2 External leakage. During all testing in Section 4, the degree of external hydraulic system leakage shall not exceed a Class 3 for dust free conditions and a Class 3 D for dusty conditions of SAE J1176. External hydraulic system leakage shall not exceed a Class 3 for production trucks under normal operation.

3.18 Uprights and carriage assembly. Uprights shall be telescopic roller type. Rollers shall be the permanently lubricated-for-life type. The "forks shall be easily removable without removing the load backrest. A cut out shall be provided in the fork carrier that allows the lower hook of the fork to pass through. Fork thickness shall not exceed 0.5 inch at the tips and the end taper shall be not less than 14 inches long.

3.18.1 Load backrest. A metal backrest, removable without disturbing any other component, shall be provided. There shall be no protruding bolts or appendages beyond the side plane of the load backrest. In conjunction with the fork and hanger design, it shall provide a vertical rear guard 42 to 46 inches high, measured from the load carrying surface of the forks, and equal to the width of the carriage, or sideshift if equipped. To allow stacking loads close to the ceiling, the upper 20 inches of the load backrest on the Type 2 trucks shall be removable and replaceable without the use of tools. It shall be secured with quick release pins when installed. Load securing rings or eyes shall be provided on both left and right sides of

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@ backrest and located not more than 12 inches from top of assembly. Safety factor for load securing devices shall be not less than 3 to 1, based on yield strength, assuming that each device is loaded equally. Ring or eye openings shall be determined using MIL-STD-209. The load securing device, when not in use, shall not extend forward of any surface on front face of backrest. The backrest assembly/ load securing device combination shall be capable of withstanding a horizontal force (at load securing devices) equal to three times the truck capacity without failure or permanent deformation of carriage assembly. Direction of force is forward.

3.19 Steering. Power Steering shall be furnished. With no load on the truck, engine idle speed and truck not moving, the tangential force on the steering wheel required to cause the truck steering wheels to turn shall not exceed 10 pounds.

3.20 Service brakes. Hydraulically actuated, self adjusting service brakes shall be provided. Service brake performance shall comply with ANSI 1356. I, except not more than 80 pounds of force on the brake pedal shall be required to meet the stopping distance and drawbar drag test.

3.21 Parking Brake. A parking brake shall be provided to lock a minimum of two wheels when brake is applied. The braking capability shall conform to ANSI B56.1. No more than 35 pounds of force shall be required to apply the parking brake.

3.22 Towing device. A pin type towing device shall be provided in the center of the rear of the truck.

3.23 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions, that enable the truck to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium, shall be provided. Slinging provisions shall conform to MIL-STD-209 and tiedown provisions shall be in accordance with MIL-STD-1791 air transportability requirements. If practical, the slinging and the tiedown points shall be the same. The notation “LIFT HERE” or "TIEDOWN” shall be stenciled in the color marking as specified (see 3.36), near each lift and tiedown point. The truck shall also be furnished with a data plate in accordance with MIL-STD-1791 which indicates the center-of-gravity location (both horizontally from front axle and vertically from ground level) in the air transport mode and tiedown instructions for transport aboard the C- I 30 and C-141 aircraft.

3.24 Battery. Unless type A winterization is specified (see 3.39.1), trucks shall be furnished with maintenance-free type battery(s). Electrical system voltage shall be 12 volts. Battery(s) shall be fully charged when the truck leaves the contractor’s facility.

3.24.1 Electrical disconnect switch. A manually operated switch shall be installed in the battery and charging circuits to permit quick disconnect of the electrical system. The switch shall be located within reach of the seated operator.

3.25 Controls and instrumentation. All load handling control symbolic markings shall be in accordance with B56.11.3. All control symbolic markings for the forklift shall be marked in accordance with SAE J1362 and SAE J898.

3.25.1 Instruments. An ammeter, tachometer, engine oil pressure gauge, engine coolant temperature gauge, transmission temperature gauge and fuel gage (except for LPG powered trucks) shall be provided. All gauges shall be identified, designed and located in accordance with SAE J209. All gauges shall be illuminated, either directly or indirectly, for night operation.

3.25.2 Hour meter. An electrically operated hour meter with the capability to register the number of engine operating hours up to a minimum of 9999 hours shall be provided. The hour meter shall be accessible for reading without removing any component of the truck. The hourmeter does not have to be lighted.

3.25.3 Ignition switch. A key operated ignition switch shall be furnished. It shall have not less than * three positions; “off”, “on” and “start”.

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3.26 Lights. Each truck shall be equipped with two forward facing, sealed beam floodlamps which shall conform to SAE J593 paragraph 46, Trade No. 4478. They shall be shock mounted in an elastomer -ring housing, one on left and one on right of overhead guard. A third identical floodlight shall be aft facing and mounted on, or near, the rear of the truck. Directional adjustment range of these floodlamps shall be a minimum of 45 degrees above and below the horizontal plane. Horizontal adjustment range of these floodlamps shall be a minimum of 180 degrees. Floodlamps shall be protected against damage by means of position or a guard. One combination stoplight and tail lamp shall be provided and shall conform to SAE J759. The tail lamp shall be recessed or mounted on the rear within the plan outline of the truck, and shall be protected against damage by a guard. An on-off switch shall be provided for the floodlamps and tail lamp. The switch shall be within access of the seated operator.

3.27 Fork control.

3.27.1 Side shift. Trucks shall be equipped with a side shift mechanism (see 3.4). The mechanism shall be hydraulically operated to move the forks a minimum of 12 inches in the front position with rated loads.

3.27.2 Spacing. Type 1 trucks shall have manually adjusted fork spacing. Type 2 trucks shall have hydraulically powered fork positioning controlled by the seated operator.

3.28 Operators overhead guard. An overhead guard meeting ANSI B56.1, including falling object protective structure (FOPS), shall be furnished. Guard height, measured from ground level to the top side of the guard, shall not exceed 88 inches. Guard design shall not interfere with the operation of the truck, nor with normal movements of the operator when entering, leaving or operating the truck. The overhead guard shall remain firmly secured to its support brackets under all operating conditions specified herein.

3.29 Seat. The seat shall conform to the requirements of SAE J899.

3.30 Tires and tire loadings. Tire loadings shall not exceed the values specified by the tire manufacturer and the Tire and Rim Association Yearbook. The truck shall be equipped with solid pneumatic tires. Tires shall be not more than 12 months old on date of acceptance by the Government.

3.31 Rearview mirror. The truck shall be equipped with the rearview mirror mounted on the left hand side within the operator's field of view.

3.32 Performance.

3.32.1 Exhaust emission requirements for clean burn diesel engine. When the diesel engine is specified, its exhaust emission constituents shall not exceed the limits set forth below when tested in accordance with applicable exhaust emission test paragraph of Section 4. These emission tests shall be conducted with a fuel delivery system identical to and calibrated to give the same fuel delivery rate as the production engine used to demonstrate conformance to the endurance requirement (see 4.6.33) and to the cold start requirement (see 3.5.1). The diesel engine as provided by the manufacturer shall be capable of satisfying the exhaust emission requirements without the use of a catalytic convertor, trap oxidizer, water scrubber or other type of exhaust gas processor with the exception that a mechanical spark arrestor in accordance with 3.14.6 shall be provided. However, the spark arrestor shall not be installed during emission testing.

3.32.1.1 Brake specific exhaust emission of hydrocarbons (HC). carbon monoxide (CO) and nitric oxides (NOx). When tested in accordance with 4.6.35.4.1, the brake specific exhaust emission of HC shall not exceed 0.5 g/bhp-hr; of CO shall not exceed 0.5 g/bhp-hr; and of NOx shall not exceed 6.0 g/bhp-hr.

3.32.1.2 Total suspended particulate (TSP). When tested in accordance with 4.6.35.4.2, the TSP shall not exceed 15 g/hr for 3, 4, 10, and 11 modes of the 13 mode Federal Test Procedure.

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3.32.1.3 Steady state smoke opacity (SSSO). When tested in accordance with 4.6.35.4.3, the SSSO shall not exceed 5 percent for any mode of the 13 mode Federal Test Procedure excluding the high load modes of 5, 6, 8, and 9.

3.32.2 Slope ascension, forward direction. Trucks shall be capable of ascending a 15 percent grade on a dry concrete surface with and without rated load. Trucks shall be able to accelerate from a dead stop on this slope when carrying rated load.

3.32.3 Upright tilt. When unloaded, truck mast shall have at least 3.0 degrees of forward tilt, and not less than 6.0 degrees of rear tilt.

3.32.4 Collapsed mast height. The collapsed mast height of a truck with no load on the forks, measured from the ground to the top of the uprights in a vertical position shall not exceed 88 inches.

3.32.5 Maximum fork height. The fork height of a truck with rated load, with the uprights fully extended, and the forks horizontal, measured from the ground to the load carrying surface of the forks at the capacity load center, shall be not less than 152 inches.

3.32.6 Free lift height. Height forks will lift without increasing the specified collapsed mast height, shall not be less than 45 inches.

3.32.7 Fork size and spacing. Fork size and spacing shall be as follows:

a. Type 1 fork size:

(1) Thickness: 1.8125 inch maximum.

(2) Width: 5.0625 inch maximum.

(3) Length: 42 inches (+0, -1 inch)

b. Type 1 and 2 fork spacing; measured to the outside of the forks:

(1) Closed position: 13 inches maximum.

(2) Open position: 37 inches minimum.

c. Type 2 fork size:

(1) Thickness: 2.0625 inch maximum.

(2) Width: 6.0625 inch maximum.

(3) Length: 42 inches (+0, -1 inch)

3.32.8 Lifting speed. Lift speed with rated load on the forks shall not be less than 55 feet per minute over the entire distance from ground level to maximum fork height.

3.32.9 Lowering speed. Speed of lowering of unloaded forks shall not be less than 40 feet per minute over the entire distance from maximum fork height to ground level. Speed of lowering forks with rated load shall be not more than $0 feet per minute over the entire distance from maximum fork height to ground level.

3.32.10 Right angle turn. Trucks, when carrying rated load, shall be capable of backing through a turn in either direction between two parallel walls no further apart than 175 inches for Type 1 and 185 inches for Type 2.

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3.32.11 Speed. While carrying the rated load on a level surface, trucks shall be capable of attaining a speed of at least 8.0 mph in both forward and reverse directions.

3.32.12 Acceleration. From a standing start, the truck when carrying rated load shall have an acceleration in the forward direction of 22 feet in not more than 5 seconds and 44 feet in not more than 8 seconds.

3.32.13 Underclearance. The truck with rated load shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 15 percent slope, and over the obstacle course described in Test Method No. 1.

3.32.14 Truck operations. The truck shalt complete not less than 3 circuits per hour, for a cumulative total of 135 hours, when tested as specified in 4.6.33. Only routine maintenance as prescribed by the manufacturer is allowed and only at the end of each operational period. The truck shall show no signs of unusual or detrimental wear. No failure shall occur during the 135 hour operational test. A failure is defined as any malfunction that cannot be corrected within 30 minutes by adjustment, repair or replacement, which may cause cessation of operation, degradation of performance capabilities below specified levels, damage to the truck by continued operation, or personnel safety hazard.

3.32.15 Fork visibility. With no load on the forks the seated operator shall be able to see at least one fork tip at all lift heights, fork spacings, and mast pivoting positions. The operator must remain seated, but may lean if necessary, to see the tip.

3.32.16 Parking brake. The parking brake shall hold the truck with rated load on a 15% grade in both the forward and reverse direction.

3.32.17 Drift. The lift assembly shall be capable of holding the rated load at maximum lift height for 10 minutes with not more than 0.5 inch of vertical load lowering and not more than 1.0 degrees of mast rotation.

3.32.18 Stability. The truck shall meet the “forward stacking”, “forward travel”, “lateral stacking”, and “lateral travel” stability requirements of ANSI B56.1.

3.33 Noise limits. The noise produced by the forklift shall conform with MIL-STD-1474 requirements when tested in accordance with 4.6.32. The provisions of MIL-STD-1474, paragraphs 4.3 and 4.4, shall be provided if and only if, MIL-STD-1474, paragraph 5.1.1.2 procedures have been accomplished, pursued and documented to the satisfaction of the procuring activity. Written permission to exceed the 84 dB (A) limit shall be obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, paragraph 4.3 and be readable at the distance indicated on the sign.

3.34 Identification marking. Each truck shall have an identified plate containing the following information: Ownership designation, nomenclature, model, contract number, truck serial number, capacity assigned registration number, gross vehicle weight (GVW), national stock number (NSN), operating weight, warranty expiration month and year, cube, date shipped, and manufacturer. The forklift shall also be equipped with instructions, warnings and caution plates prominently located and describing any special or important procedure to be followed in operation, loading and servicing the forklift or its components. All plates shall conform to MIL-P-514, Type III, Composition C (GG-P-455) Type 1, Grade A, Class 1, unless otherwise specified. The identifying data shall be stamped in letters in accordance with MIL-P-514. All plates shall be securely attached to the truck with bolts, rivets or screws and shall be furnished and mounted by the contractor.

3.35 Painting. Ail external surfaces and all surfaces normally painted by the truck manufacturer shall be prepared and primed using material compatible with the final paint. Unless otherwise specified (see 6.2), the truck shall be painted with Polyurethane Paint, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be DuPont Imron, or equal. Or equal is defined as having performance identical, or superior to, DuPont Imron when subjected to the following tests: Paragraph 4.3.17 and 4.3.23 of MIL-C-46168 and 4.8.6, 4.8.10, and 4.8.12 of MIL-C-83286. All exterior trim (wheels,

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radiator grill, rims on lights, etc) shall be the same type and color as the exterior paint. The paint finish shall show no evidence of uneven application, curtains, runs, snags, orange peel, lack of adhesion, or other defects.

3.36 Truck marking. Unless otherwise specified (see 6.2), markings on truck shall be painted black. Characters shall be block-type capitals and Arabic numerals. Location and character size shall be as follows:

Information	Example	Location	Min. Character Size (Ht.)
Agency	US	On each side and rear	1.5 in.
Registration No.	10D315	On each side and rear	1.5 in.
Capacity	4000 lb	On each side of mast	2.0 in.
Safety	No Smoking	On rear of mast	l.0 in.
	No Riders	On rear of mast	1.0 in.
Type fuel	Diesel	Near fuel tank fitter	1.0 in.
Lifting/Tiedown Points	Lift/Tie-down	Near lift/tie-ciown points	1.0 in.

3.37 Electromagnetic interference (EMI). The truck shall comply with the EMI requirements of MIL-STD-461B, Class Cl, Group II, Figure 8-1, except the emission limits for UM03 (Figure 8-1) shall be relaxed by 20dB for all frequencies.

3.38 Cold weather operation. The forklift shall be operable by personnel wearing the following ensemble:

a. Common winter parka.

b. Heavy gloves, such as ski or snowmobile gloves.

c. Insulated, high-top, hunting boots.

The operator shall be capable of actuating all controls (i.e., levers, switches, pedals, knobs, and handles) without interference caused by the ensemble that is detrimental to the operation of the forklift.

3.39 Winterization. When specified (see 6.2), the truck shall be winterized as follows:

3.39.1 Type A. Type A Winterization shall protect to -65 degrees F, and shall consist of the following components:

a. Power plant heater. Coolant, engine oil and battery heaters shall be provided. Heaters shall operate on 110-volt, alternating current, and shall be wired through a junction block to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and grounding vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation, of connecting cable, shall withstand normal operating stresses in low ambient air temperature (down to minus 65 degrees F) without loss of dielectric capacity. Ail heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

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(1) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. A coolant circulating pump, driven by a 11 0-volt alternating current motor, shall be provided.

(2) Immersion type engine oil heater, 300 watt minimum rating with 190 to 195 degrees F thermostat, shall be installed in oil pan.

b. Battery. Heavy duty (lead-acid) battery shall be furnished. The battery shall be heated by an electrical heater. Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10 degrees F during vehicle exposure in ambient temperature as low as minus 65 degrees F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than +80 degrees F.

c. Cab. A cab that is either removable from, or integral with, the overhead guard shall be provided. Thermal insulation shall be installed on the floor, cab side walls and cab ceiling, with washable covering over all insulation. Boots are required for insulation to close all openings where attachments and controls enter the cab. There shall be windshield wipers on the front windshield, and rear and roof windows.

d. Thermostat. Install a 170 to 190 degrees F engine thermostat.

e. Personnel cab heater with defroster. A hot water heater and defroster shall be installed. The heater shall be of sufficient capacity to maintain a temperature of +40 degrees F at cab floor level in an ambient temperature of -40 degrees F. Use may be made of sufficient thermal insulation material in the cab to ensure that the cab heater can meet these temperature conditions.

f. Air dryer. An air dryer shall be included if air brakes are furnished.

g. Arctic type lubricants. Engines, gearboxes, transmissions and other reservoirs shall be compatible with arctic type lubricants.

h. Antifreeze. Antifreeze shall comply with either Military Specifications MIL-A-11755 or MIL-A-46153 or shall be standard commercial arctic type. Coolant liquid shall be protected to at least -65 degrees F.

3.39.2 Type C. Type C Winterization shall protect to -25 degrees F and shall consist of the following components:

a. Cab. A cab conforming to 3.39.1.c. shall be included.

b. Antifreeze. Antifreeze shall comply with either Military Specifications MIL-A-11755 or MIL-A-461 53 or shall be standard commercial type. Coolant liquid shall be protected to -25 degrees F or lower temperature.

c. Thermostat. Install a 170 to 190 degrees F engine thermostat.

d. Air dryer. Air dryer shall be included if air brakes are furnished.

e. Personnel cab heater with defroster. A hot water heater and defroster shall be installed. The heater shall have sufficient capacity to maintain a temperature of +40 degrees F at cab floor level in an ambient temperature of -25 degrees.

3.40 Welding. All welds shall be smooth, continuous, with complete heat penetration and shall transmit stress without permanent deformation or failure when the truck is subjected to normal service

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and the testing in Section 4 of this purchase description. The contractor's welders, welding operators and welding procedures shall be qualified to meet the requirements of MIL-STD-248, or the applicable codes of American Welding Society (AWS) or American Society of Mechanical Engineers (AMSE).

3.41 Drain holes. Structures shall be designed to avoid pockets where debris and water might accumulate. Ail structural configurations shall allow water run off, or shall have adequately sized and placed drain holes.

3.42 Workmanship. The workmanship shall be high quality when compared to industry standards. There shall be no rough surfaces or edges on machined or frame cut pieces. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist.

4. QUALITY ASSURANCE Provisions

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. 'The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet ail requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined and tested in accordance with the requirements of referenced specification and standards.

4.3 Classification of inspections, evaluations, and demonstrations. inspections, evacuations, testing, and demonstrations shall be classified as follows:

a. First article truck inspection (see 4.4).

b. Production run-in (see 4.7).

c. inspection of packaging (see 4.8).

cf. User test (see 4.9).

e. Random sample test (see 4.1 0).

4.4 First article truck inspection. Prior to production the manufacturer shall furnish one truck produced from production tooling for first article evacuation and test performed by the contractor. Prior to examination and test, the following shall be performed.

a. Service the truck with the oils, greases, and fuel, specified herein and designated for use in the ambient temperature at which the test will be conducted.

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b. Conduct run-in as prescribed by the manufacturer.

c. Set the hydraulic system relief pressure as specified by manufacturer and record. Any change made to this except during 4.6.29 shall constitute failure of all tests.

d. Furnish a schedule of maintenance to be followed during all testing of the first-produced trucks. A list of special tools required to perform this maintenance shall also be provided.

4.4.1 Examination. 'Prior to -testing, the first article truck" shall be examined and compared with the requirements of Section 3 of this document.

4.4.2 Tests. Upon successful completion of examination specified in 4.4.1, the truck shall be subjected to the demonstrations and tests specified in 4.6. All of these tests except Cold Start (see 4.6.34) and Diesel Exhaust Emissions (see 4.6.35) shall be conducted using JP-8, Jet-A or Jet-Al as the fuel.

4.5 Inspection failure. Failure of the first article truck to meet any requirements during examination (see 4.4.1) or occurrence of any of the following conditions during the testing (see 4.6, 4.7, 4.9 and 4.1 0) shall constitute failure of the first article evaluation and shall be cause for rejection:

a. Inability to meet specified performance requirements or inability to complete any test.

b. Permanent deformation, overheating, malfunction or leakage of fuel, or lubricants, or leakage of hydraulic fluid in excess of Class 3 or 3D of SAE J1176.

c. Any binding, jerking or uneven operation of any component or system.

d. Any wearing, galling, gouging or other removal of material that, in the opinion of the Government Contracting officer, will cause premature failure.

4.6 Demonstration and tests.

4.6.1 Demonstration and test conditions. Unless otherwise specified herein, demonstrations and tests shall be conducted at the ambient temperature and climatic conditions existing at the place of demonstration and test. Only that maintenance scheduled by the contractor and submitted as a maintenance schedule prior to commencement of the demonstration and test shall be performed during the demonstration or testing.

4.6.2 Rated load. Rated load shall be established with a load equivalent to an unrestrained cube with overall dimensions twice the load center dimensions and whose center of gravity is located at the geometrical center of the cube. The tolerance for weight shall be plus 2 percent and minus 0 percent. The cube for Type 1 shall weigh 4000 pounds and Type 2 shall weigh 6200 pounds.

4.6.2.1 Load center. The load center shall be defined as the horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and vertical load engaging faces of the forks to the center of gravity of the load. The tolerances for dimensions shall be plus or minus 1 percent.

4.6.3 Maximum fork height, collapsed mast height, overhead guard height, underclearance and freelift height. With no load and mast in true vertical position, measure and record collapsed mast height, free lift height, overhead guard height; with rated load and mast in true vertical position, measure and record maximum fork height and underclearance. Nonconformance to 3.8.4, 3.32.4, 3.32.5, 3,32.6 or 3.32.13 shall be cause for rejection.

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4.6.4 Lifting speed and lowering speed. Measure and record distance in inches from floor to top surface of forks in fully lowered position. Record time in seconds required to raise rated load to maximum fork height. Record the difference between maximum fork height and the height of the forks in lowered position. Record time in seconds to lower rated load at maximum speed (full open valve) to 3 or 4 foot height, load to be abruptly stopped at the 3 to 4 foot height. Repeat the raising and lowering three times. Remove load and raise empty forks to maximum lift height. Record time in seconds required to lower forks to lowered position. Nonconformance to 3.32.8 or 3.32.9 shall be cause for rejection.

4.6.5 Right angle turn. Starting with the rated load in the carry position, but with the mast vertical (see 6.5), and the front of the load against a wall (or a line on the floor) and the forklift perpendicular to the wall. Back the truck through a right angle (90 degree) turn. Mark the point where the outside envelope of the forklift is farthest from the wall during the turn. Measure the perpendicular distance from this point to the wall. If it exceeds the amount specified in 3.32.10, this shall be cause for rejection. Repeat the procedure, turning in the opposite direction, again not exceeding the amount specified in 3.32.10.

4.6.6 Travel speed. Operate the truck at governed speed on a level surface with rated load in carry position. Truck shall attain speed specified in both directions. The speed shall be the average of three runs in each direction. Inability of the truck to attain the minimum speed as specified in 3.32.11, failure of any component, over-heating, malfunctioning, or leaking of fluid shall be cause for rejection.

4.6.7 Acceleration test. Run acceleration test as required in 3.32.12. Nonconformance to 3.32.12 shall be cause for rejection.

4.6.8 Slope ascension. Drive truck forward with rated load along a horizontal approach and onto a ramp or slope of grade specified in 3.32.2 and bring to a complete stop using the service brakes. Start from a dead stop and proceed up to the top of the ramp. Remove capacity load and repeat above test. Use a template or other device to verify that the truck can go from a 15% grade onto a horizontal surface. Nonconformance to 3.32.2 shall be cause for rejection.

4.6.9 Tilt. Place truck without load on flat level surface. Raise forks to an elevation of about 2 feet. Tilt mast to full forward position and record angle of tilt, measured from the front surface of the outer most channel of the mast. Tilt mast to full rearward position and record angle of tilt measured from the same surface. Nonconformance to 3.32.3 shall be cause for rejection.

4.6.10 Fail-safe hydraulic lift system. Raise the rated load four to 10 feet above ground level with the mast approximately vertical. Measure vertical distance to one fork tip. Disconnect a hydraulic line that supplies pressure to the lift system to simulate a line failure. When fluid stops draining, remeasure vertically to same point on fork tip. A drop of more than 0.5 inch shall be cause for rejection. Measure the mast tilt. Disconnect the hydraulic lines that supply pressure in both directions, to all tilt cylinders. When fluid stops draining, remeasure mast tilt. A change of more than one degree shall be cause for rejection. Lower the forks to ground level using the override system.

4.6.11 Drift. Exercise the lift and tilt functions to heat the hydraulic fluid to operating temperature. Raise the rated load to maximum lift height with mast approximately vertical. Measure vertical distance from ground to bottom of load at the 24 inch load center. Measure mast tilt from vertical. Shut off engine and let sit for 10 minutes. Remeasure vertical distance to same point on bottom of load. Remeasure mast tilt. Nonconformance with the drift requirements of 3.32.17 shall be cause for rejection.

4.6.12 Stability. Perform applicable stability tests of ANSI B56.1 specified in 3.32.18. Inability to pass the ANSI tests shall be cause for rejection.

4.6.13 Service brakes. Perform the drawbar and stopping distance service braking tests specified in ANSI B56.1 as modified by 3.20. Nonconformance shall be cause for rejection.

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4.6.14 Parking brake. Drive the truck forward, while carrying rated load, onto a 15% ramp. Measure the force at the handle required to fully apply the parking brake. Using only the parking brake, hold the truck for 2 minutes. Repeat this test, except test with the truck up the ramp in reverse. The truck may be driven up on the ramp in reverse with no load on the forks and the load applied with the truck stationary on the ramp. Any movement of the truck while the parking brake is applied shall be cause for rejection.

4.6.15 Starter disconnect. Start the truck, energizing the starter switch. Allow the engine to run for a minimum of 15 seconds. Re-energize the starter switch while the engine is running. Any evidence of starter engagement while engine is running shall be cause for rejection.

4.6.16 Transmission neutral switch. With the engine not running, place transmission selector in lowest forward gear and energize starter switch. Repeat for all forward and reverse gears. Any evidence of starter engagement while transmission is in gear shall be cause for rejection.

4.6.17 Slinging and tiedown provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD-209. Test the tiedown provisions by applying the ultimate load for the time period specified in MIL-STD-209. Nonconformance to 3.23 shall constitute failure of this test.

4.6.18 Tire loading. While transporting rated capacity loads the tires shall not exceed the values listed in the Tire and Rim Yearbook, as applicable.

4.6.19 Electromagnetic interference (EMI). Test the truck to verify that it meets the EM I requirements of 3.37. Failure to meet these requirements shall be cause for rejection.

4.6.20 Fork tines. Measure the length, width, and thickness of the fork tine to determine compliance with 3.32.7.

4.6.21 Sideshift and fork spacing. With no load on the forks sideshift to extreme left position. Sideshift to extreme right and measure total travel. Move forks together to the most closed position. Measure to outside of forks. Move forks to most spread apart position and measure to outside of forks. Nonconformance with 3.27 and 3.32.7 shall be cause for rejection.

4.6.22 Fork visibility. With no load on the forks, vary the lift height, side shifting, fork spacing and mast pivoting to verify compliance to the visibility requirements of 3.32.15.

4.6.23 Steering. With no load on the forks, engine at idle speed and truck stationary, measure tangential force on the steering wheel required to cause the steering wheels to turn. Failure to meet the requirements of 3.19 shall be cause for rejection.

4.6.24 Axle weight. Weigh front and rear axle with the forklift fully operational, with the cab installed (if applicable), and at least 0.25 tank of fuel. No axle shall weigh more than 13,000 pounds.

4.6.25 Overload. Place truck on level surface and support it with blocks under the axle or frame to relieve the load on the tires. Secure the truck to the floor in a manner to compensate for the additional test load. The cylinders may be hydraulically locked for this test. With the mast in straight-ahead position gradually apply load to the forks until three times the rated capacity (12,000 pounds for Type 1 and 18,600 pounds for Type 2, at 24 inch horizontal and vertical load center) is on forks. Leave this load on the forks for at least 10 minutes. Remove load and inspect truck structure, frame, mast assembly and hydraulic system for deformation, cracks, broken welds, hydraulic system leaks, etc. inspect forks at least six inches each side of heel by magnetic particle or dye penetrant method.

4.6.26 Overhead guard. Perform the falling object protective structure (FOPS) test of ANSI B56.1. Failure to meet the criteria of this test shall be cause for rejection.

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4.6.27 Load securing device. Apply a total forward, horizontal load of 3 times rated capacity (12,000 pounds for Type 1 and 18,600 pounds for Type 2) divided equally between ail load securing devices (see 3.18.1). The entire test load does not have to be applied at one time, the test for each device may be applied independently. Any deformation, broken welds, etc. shall be cause for rejection.

4.6.28 Fork lifting mechanism. The fork lifting mechanism shall be operated through 100 complete continuous cycles (from ground level to maximum lift height and back to ground level) with no load and 200 cycles with rated load on the forks. The test shall be performed at a rate of not less than two cycles per minute. Any failures to lift mechanism components overheating of engine coolant system or hydraulic system, inability to complete the test or any other failure shall be cause for rejection.

4.6.29 Fork lifting overload. With 125 percent of the rated load (5,000 pounds for Type 1 and 7750 pounds for Type 2 at 24 inch horizontal and vertical load center) on the forks and with the truck tied down to prevent overturning, the lifting mechanism shall be operated through 25 complete continuous cycles. From ground level to maximum lift height and back to ground level shall be considered one cycle. Side shift the 125 percent load from extreme right to extreme left and back to extreme right. This shall be considered one cycle. Perform at least 25 continuous cycles. Both the lifting and side shifting shall be performed at a rate of not less than two cycles per minute. The occurrence of any condition listed in 4.5 shall be cause for rejection.

4.6.30 Fork spating mechanism. For Type 2 trucks, the fork spacing shall be operated with no load on the forks through 100 complete, continuous cycles. A cycle shall consist of starting at closed position, going to open position and returning to closed position. Nonconformance with 4.5 shall be cause for rejection.

4.6.31 Fork side shift. With the rated loads on the forks the side shift shall be operated through 100 complete, continuous cycles. A cycle shall consist of starting at extreme right, side shift to extreme left, then side shift back to extreme right. Nonconformance with 4.5 shall be cause for rejection.

4.6.32 Sound level measurements.

4.6.32.1 Operator's station. Measure the maximum steady state noise levels of the truck in accordance with procedures of MIL-STD-1474.

4.6.32.2 Exterior. Measure the maximum steady state noise level in accordance with MIL-STD-1474 for exterior noise for construction and materials handling equipment. Conformance or nonconformance shall be evaluated per 3.33.

4.6.33 Truck operations. Perform Test Method No. 1, Truck Operations. Nonconformance with 3.32.1 and 3.32.14 shall constitute failure of this test.

4.6.34 Cold starting. Place the truck in an ambient temperature of -25 degrees F, or cooler, until all components are stabilized at the ambient temperature. The engine shall start within five minutes and run smoothly at idle speed without continued engine control adjustment within 15 minutes. At this time all travel, braking, and load handling mechanisms, shall be operated. The forks shall be raised to maximum lift height allowed by the test chamber, and sideshifted and adjusted laterally between extreme positions. Starting aides such as the ether injection or glow plug furnished with the truck may be used to assist starting; however, no external power or heat source shall be used. The test chamber shall maintain the ambient temperature as close to -25°F as possible during the entire test. Failure to meet any of these requirements shall be cause for rejection.

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4.6.35 Diesel exhaust emissions.

.-4.6 .35.1 Diesel exhaust emission tests. Diesel exhaust emission testing shall not commence until the contractor has demonstrated conformance to the endurance and cold start requirements.

4.6.35.2 Selection/verification preparation of diesel engine for exhaust emission tests. Diesel exhaust emissions test shall be conducted on two engines supplied by the engine manufacturer. Each engine shall be equipped for dynamometer testing with production accessories (air cleaner, alternator, fan, etc.) as tested in the endurance and cold start tests. The fuel delivery system of the two test engines shall be identical and their adjustment and fuel delivery rates shall be determined and verified to be within 1 percent of those of the engine used in the endurance and cold start tests. If the fuel delivery systems are not identical and adjusted and calibrated to be within 1 percent of each other, this shall be cause to prohibit commencement of further emission testing until corrected. Each of the test engines shall then be run-in for at least 80 hours. During the run-in the engines shall be subjected to an even distribution of the loads and speeds from the 13 modes of the Federal test procedure. After the run-in is complete, the horsepower of each of the emission test engines shall be determined. Each engine shall be capable of developing horsepower within 5 percent of the rated horsepower at rated speed as published by the engine manufacturer for this application. Failure of the test engines to develop sufficient horse power until corrected shall be cause to prohibit further testing.

4.6.35.3. General emission test conditions. Each emission test specified herein shall be performed two times with each emission test engine and the results averaged to obtain the value which is compared to appropriate specification requirements. All emission testing shall be conducted using a D2 diesel control fuel as specified by 40 CFR Parts 81-99. The 80 hour run-in may be conducted with the contractor's choice of diesel fuel.

4.6.35.4 Emission test procedures.

4.6.35.4.1 Emission test procedure for HC, Co, and NOx. Determine the emissions rate of HC, CO and NOx in accordance with the steady-state 13 mode Federal Test Procedure (FTP), for new diesel heavy duty engines as specified by Title 40 Code of Federal Regulations, subpart D. Brake specific values after being averaged in accordance with 4.7.20.3 or more than 0.5 g/bhp-hr HC, of 5.0 g/bhp-hr CO, and 6.0 g/ bhp-hr NOx as specified in 3.32.1.1 shall constitute failure of this test.

4.6.35.4.2 Total suspended particulate (TSP) emission test procedure. The TSP emission measurement system shall be in accordance with procedures outlined by the Federal Register Volume 40, Part 86. Measurement shall be made, however, with the engine at steady state modes taken from the 13 mode Federal Test Procedure. Testing for TSP 3, 4, 10, and 11 modes of the 13 mode test shall be conducted. Average TSP emission rates exceeding 15 g/hr at any mode as specified (see 3.32,1 .2) shall constitute failure of this test.

4.6.35.4.3 Steady state smoke opacity test and procedures. Steady state smoke opacity levels shall be determined using the smoke measurement system specified by Title 40, Code of Federal Regulations Subpart I for new diesel heavy duty engines. Instead of the 3 modes specified in subpart I the engine modes to be tested shall be taken from the steady state 13 mode Federal Test Procedures (FTP) as specified by Title 40, Code of Federal Regulations Subpart D for new gasoline fueled and diesel heavy duty engines. The high load modes 5, 6,8, and 9 shall be excluded from the smoke test. Measurement of smoke opacity at all of the remaining of the 13 mode FTP test shall be conducted. Average smoke opacity values exceeding 5 percent at any mode as specified in 3.32.1.3 shall constitute failure of this test.

4.6.36 Cod weather operation. With the operator wearing the ensemble listed in 3.38, verify by operating the truck that the operator can actuate all controls. Any interference caused by the ensemble that is detrimental to the operation of the truck shall be cause for rejection.

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4.7 Production run-in. Each truck produced on the contract shall be subjected to a production run-in and servicing as specified herein. Inability to complete the run-in, overheating, failure of any components, malfunction of any control, or evidence of leakage of fluids in excess of Class 3 in accordance with SAE J1176 shall be cause for rejection of the truck by the Government until such deficiencies are corrected.

4.7.1 Travel. While carrying rated load, demonstrate starting, ascending and descending a 15 percent ramp, steering, inching and travel. Operate all lights, horn and brake system not less than two times during this demonstration.

4.7.2 Mast assembly. Demonstrate ability to lift rated load to full lift height at full lift speed and lower load at full opened valve without exceeding maximum lowering speed. Demonstrate ability to tilt mast full forward and full rearward without rated load on the forks. Demonstrate ability to side shift rated load throughout full range of side shift. Demonstrate ability to feather controls of all hydraulic functions. Each operation shall be performed at least three times.

4.7.3 Servicing and adjustment. Prior to delivery, the manufacturer shall service and adjust each truck for proper operational use, including at least the following:

a. Adjustment of mast assembly.

b. Adjustment of engine, transmission, and power train.

c. Adjustment of brake system.

d. Adjustment of steering.

e. Continuity check of electrical system, including filling and charging of battery unless otherwise specified in the preservation and packing requirement.

f. Check torque on all wheel lug nuts and proper inflation of tires.

g. Service the cooling system with antifreeze water solution.

h. Check fluid levels and fill to required level if necessary.

4.8 Inspection of packaging. The preservation, packing and marking of each forklift truck shall be inspected to verify they meet the requirements of 5.1.

4.9 Use test The User Test forklift shall be delivered to Nellis AFB NV for 45 available days (seven days a week) of user tests. The 45 available days shall be consecutive, except when interrupted for maintenance/repair downtime, as defined herein. The User Test will not begin until after completion of the first article tests, except the EM I and environmental tests, of the contractor test sample as defined by 4.3.a. Production run-in (see 4.8) and the examination of product (see 4.4.1) shall be performed on the user test sample prior to shipment of the user test sample to Nellis AFB. The user test sample shall not be shipped to Nellis AFB without prior 15 day notification to the POC. The test period shall begin the day after the vehicle arrives at Nellis AFB and shall end when 45 days of availability are complete or 60 calendar days have elapsed, whichever occurs first. If the user test sample has not been available for 45 test days after 60 calendar days have elapsed, the User Test shall be considered a failure. The vehicle shall be available for any one period of eight consecutive hours during each day of the 45 day test. The periods shall be scheduled by the Nellis AFB representative and the schedules shall be provided to the contractor at least 12 hours in advance. Any period of more than two hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day for the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to

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be supplied with the test sample. The preliminary technical data shall contain procedures for preparation/ tiedown of the forklift for air shipment. The contractor shall provide any special tools/equipment required to perform operators maintenance scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before notification to a Nellis representative, and Government personnel shall be allowed to observe their action. The contractor shall provide all forklift consumables required in the course of the test except for diesel fuel, water, compressed air and electrical power which will be furnished by Nellis AFB. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in, or will simulate, normal Air Force usage and will not exceed any performance requirements defined by Section 3 of this specification. Noncompliance with the performance requirements of this specification shall be cause for rejection. The contractor shall maintain a daily log. The log shall be signed by the Nellis AFB representative and the contractor after each day of activity. The daily log shall document, but not be limited to the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. Nellis AFB representative comments.

The contractor shall include the user test results, including daily logs of activities, in the First Article Inspection Report.

4.10 Random sample test. Any vehicle from the production line may be selected by the procuring activity at any time during the production phase of the contract to be subjected to the examination in 4.4.1 and the test described in 4.4.2. Any deficiencies found as a result of the inspection or test shall be considered prima facie evidence that all trucks already accepted are similarly deficient. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on truck previously produced and accepted under the contract, as well as on ail trucks to be produced during the remainder of the contract.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing and marking shall be in accordance with the requirements of the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Front/side loading trucks described herein are intended to operate in minimum and very narrow aisles with pallets and long loads when they are stacking, unstacking, and moving cargo in and around warehouses, loading platforms, and docks within the military supply system; also for moving cargo in and out of transport carriers such as highway trailers and vans, cargo containers, railroad cars, and cargo vessels. Diesel powered trucks will be operated in and out of earth covered ammunition storage facilities and in other uses where internal combustion powered trucks are operated. These solid fill pneumatic tired trucks are also intended for operation over paved, semiprepared and other hard surfaces for short distances.

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6.2 Ordering data. Procurement documents should specify the following:

a. Number and date of this specification.

b. Type of forklift to be furnished (see 1.2).

c. When the engine shall be other than diesel type (see 3.1).

d. When a User Test is required (see 3.2).

e. When rustproofing shall be applied (see 3.3.4),

f. When final color shall be other than as specified (see 3.35).

g. When color of markings shall be other than as specified (see 3.36).

h. Type of winterization, if any (see 3.39).

i. 'Time frame for submission of first article test report (see 6.7).

6.3 incident report. When the contractor conducts the demonstration and test specified herein, a written report within 24 hours, of any malfunction or failure during the conduct of the demonstration and test is required. As a minimum, the report shall describe components and parts affected, test and operating conditions, date of incident, hour meter reading, how detected and description of incident and corrective action taken. A minimum of one original will be furnished the pertinent contracting officer specified in the contract. Distribution of copies will be identified in the contract.

6.4 Fail-safe hydraulic lift system. A system will inherent safety features which will automatically prevent load lowering and mast tilting in the event of hydraulic line failure and provide a means of manual overriding the system to safely and slowly lower the loaded or empty carriage so that repairs can be made without incident.

6.5 Carry position. Carry position is defined as the forks in the straight ahead position with the heel of the forks 4 to 12 inches above ground, and the mast tilted to the most rearward position.

6.6 Data requirements. The following Data item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied to a contract, in order to obtain this data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Para	DID Number	DID Title	Suggested Tailoring
3.2	DI-T-4901	First Article Test Procedures	None
3.2	DI-T-23731A/T	Notification of Tests	None

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management System and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.7 First article. When the first article inspection is required, the item(s) to be tested shall be a first article sample. The contracting officer shall provide specific guidance to offerors as to the number of items to be tested as specified in 3.2. The contracting office shall also include specific instructions in

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acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids shall provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders shall not submit alternate bids unless specifically requested to do so in the solicitation. Approval of the first article truck shall not relieve the contractor of his obligation to furnish forklifts conforming to this specification.

6.8 Subject item (key word) listing.

Material handling
Pivot mast
Swing mast

Custodians:
Air Force-99

Preparing activity
Air Force-84

(Project 3930-F022)

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TEST METHOD NO. 1

TRUCK OPERATIONS

1. Demonstration course:

   (a) The test course shall be a level, improved surface laid out in accordance with Figure A-1. Station 1 and Station 3 shall be platforms that are not less than 8 feet in height and capable of supporting rated load. Station 2 shall be at ground level.

   (b) Obstacles shall be in accordance with Figure A-2.

2. Demonstration procedure:

   (a) Initially, place rated loads on platforms at Stations 1 and 3.

   (b) Pick up load from Station 1; back up truck being sure all wheels travel over obstacles; travel forward to Station 2. Before reaching Station 2, stop truck and pivot mast with load 90° to right, then pivot mast back to straight ahead position.

   (c) Deposit load to Station 2; back out; then travel forward to Station 3. Prior to engaging load at Station 3, raise mast to maximum lift height to activate relief valve. Lower forks and engage load.

   (d) Pick up load at Station 3; back out; then travel forward and deposit load at Station 1, being sure all wheels travel over obstacles.

   (e) Disengage load at Station 1; back out, being sure all wheels travel over obstacles; then travel forward to Station 2.

   (f) Pick up load at Station 2; back out; then travel forward and before reaching Station 3, stop truck, pivot load 90° to right, then pivot back to straight ahead position. Deposit load at Station 3.

   (g) Steps 2 (b) through 2 (f) shall constitute one circuit. For 50 percent of the circuits, Station 2 and Station 3 shall be interchanged so that they are in clockwise position rather than in their initial counterclockwise position. Stations shall be changed at the end of each 8-hour operating day.

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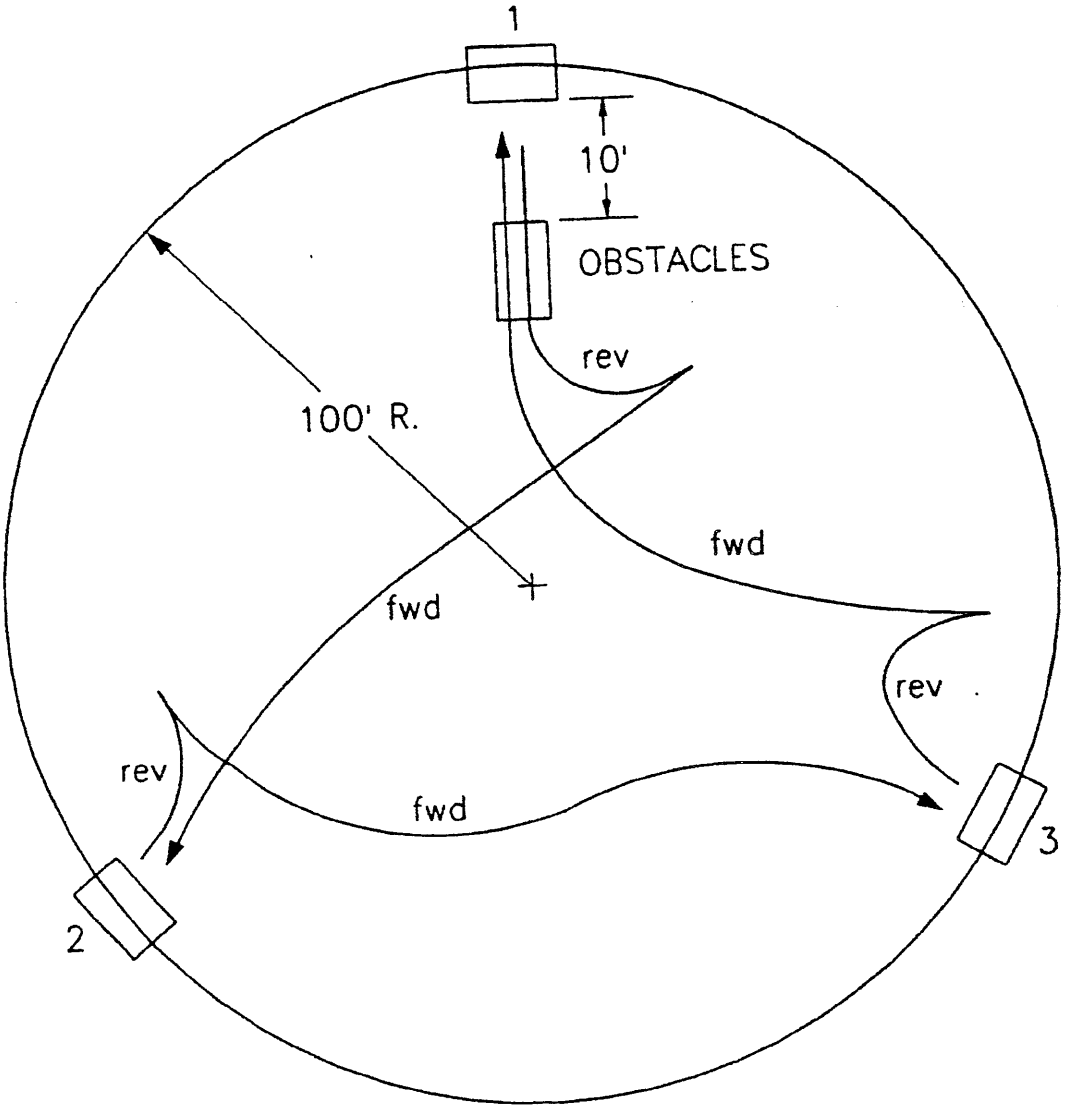
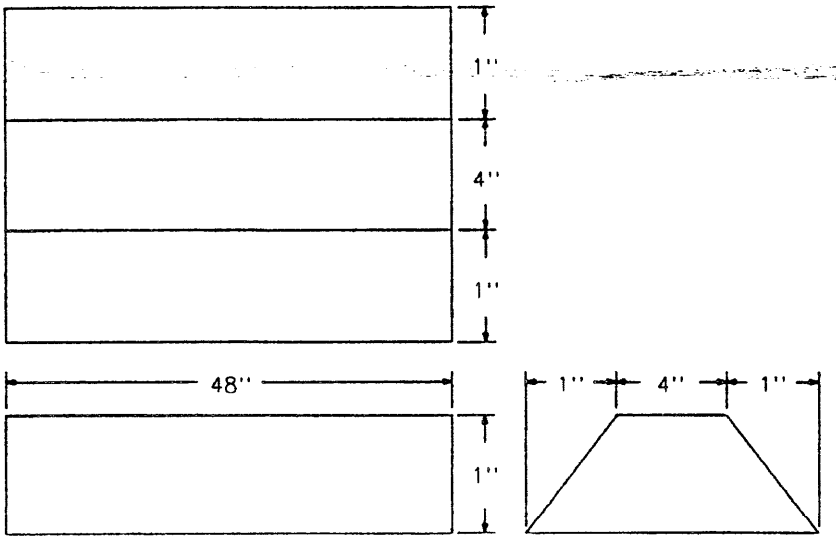


Figure A-1 TRUCK OPERATIONS COURSE

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LAYOUT OF OBSTACLE COURSE

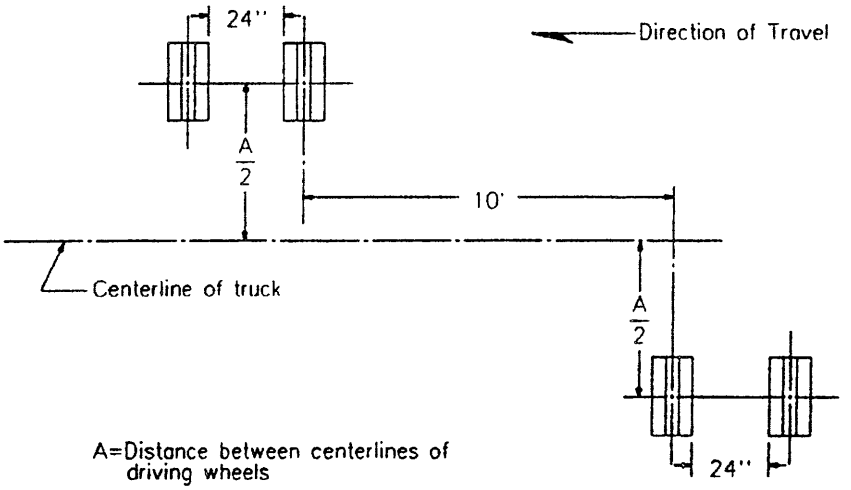


FIGURE A-2 DETAILS OF OBSTACLE CONTRUCTION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
George Cyphers	(912)926-6171	AV 468-6171
c. ADDRESS (Include Zip Code) WR-ALC/MMVRG Robins AFB, GA 31098-5609	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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